

HW # 5

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33.3-1 HUMIDITY FROM VAPOR PRESSURE

$$T_{air} = 37.8^{\circ}C$$

$$P_{total} = 101.3 \text{ kPa} = 101300 \text{ Pa}$$

$$P_A = 3.59 \text{ kPa} = 3590 \text{ Pa}$$

(a) From the STEAM TABLES @ $37.8^{\circ}C$, the vapor pressure of water is $P_{As} = 6.56 \text{ kPa abs}$ (0.951 psia). Also, we are given $P_A = 3.59 \text{ kPa}$ and $P = 101.3 \text{ kPa}$ (14.7 psia). In order to calculate humidity, we use:

$$H = \frac{\text{kg H}_2\text{O}}{\text{kg dry air}} = \frac{18.02}{28.97} \frac{P_A}{P - P_A} \quad (33.3-1)$$
$$= \frac{18.02}{28.97} \cdot \frac{3.59}{(101.3 - 3.59)} = \frac{64.69}{2830}$$

$$\therefore H = 0.02285 \text{ kg H}_2\text{O} / \text{kg dry air} \quad \checkmark$$

(b) Saturation humidity is given by:

$$H_s = \frac{18.02}{28.97} \frac{P_{As}}{P - P_{As}} \quad (33.3-2)$$
$$= \frac{18.02}{28.97} \cdot \frac{6.56}{(101.3 - 6.56)} = \frac{118.2}{2742.02}$$

$$\therefore H_s = 0.04311 \text{ kg H}_2\text{O} / \text{kg air}$$

(c) % relative humidity is given by:

$$H_R = \frac{P_A}{P_{As}} \cdot 100\% \quad (33.3-4)$$
$$= \frac{3.59}{6.56} \cdot 100\% = (0.547) 100\%$$

$$\therefore H_R = 54.73\%$$

33.3-3 Use of Humidity Chart

$$T_{air} = 65.6^{\circ}\text{C} \quad (150^{\circ}\text{F})$$

$$T_{dp} = 15.6^{\circ}\text{C} \quad (60^{\circ}\text{F})$$

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Solution: The dew point of 15.6°C is the temperature at which the given mixture is at 100% saturation. Starting at 15.6°C , we draw a vertical line (Fig 33.3-2) until it intersects the 100% humidity line, where we can read a humidity of:

$$H \approx 0.0113 \text{ kg H}_2\text{O} / \text{kg dry air.}$$

This is the actual humidity of the air @ 65.6°C , i.e. If air at 65.6°C w/ a humidity $H = 0.0113$ is cooled, its dew point will be 15.6°C .

Locating the point where $H = 0.0113$ and $T = 65.6^{\circ}\text{C}$, the percentage humidity H_p is found to be 5.3%, obtained by linear interpolation vertically between the 5% and 10% lines.

The humid heat for $H = 0.0113$ is:

$$C_s [\text{kJ/kg dry air} \cdot \text{K}] = 2.005 + 1.88 H \quad (33.3-6)$$

$$C_s = 2.005 + 1.88(0.0113)$$

$$\therefore C_s = 1.0262 \text{ kJ/kg dry air} \cdot \text{K}$$

The humid volume at 65.6°C is:

$$v_H [\text{m}^3 / \text{kg dry air}] = \frac{22.41}{273} T(\text{K}) \left(\frac{1}{28.97} + \frac{1}{18.02} H \right) \quad (33.3-7)$$
$$= \left[(2.83 \times 10^{-3}) + (4.56 \times 10^{-3})(0.0113) \right] (65.6 + 273.15)$$

$$\therefore v_H = 0.976 \text{ m}^3 / \text{kg dry air}$$

33.3-9 Dehumidification of Air

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$$\begin{array}{l} T_{\text{dry bulb}} = 37.8^\circ\text{C} \\ T_{\text{wet bulb}} = 26.7^\circ\text{C} \end{array} \left\{ \begin{array}{l} \text{cooling} \\ \text{drying} \end{array} \right. \rightarrow T_c = 15.6^\circ\text{C} \xrightarrow{\text{heat}} T_h = 23.9^\circ\text{C}$$

(a) Given that T_{db} & T_{wb} have been given, we can calculate the dew point of the initial air by investigating the Dew Point Temperature chart.

Thus, from the psychrometric chart for bone dry air, dew point for the initial air is:

$$T_{dp} \approx 24^\circ\text{C}$$

Again, reading the charts,

$$H \approx 0.020 \text{ kg H}_2\text{O/kg dry air}$$

by intersecting the 24°C vertical line w/ 100% humidity curve.

Relative humidity is given by:

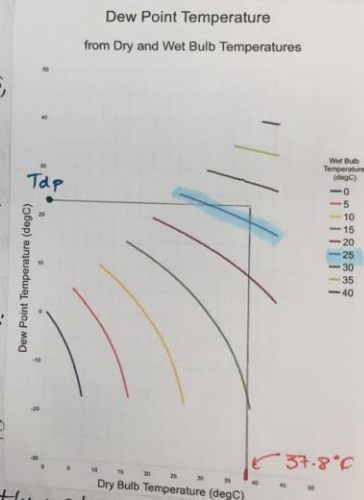
$$\% H_R = \frac{P_{H_2O}}{P_{H_2O}^*} \cdot 100\%, \text{ where}$$

• P_{H_2O} is the partial pressure of water vapor in the gas mixture

• $P_{H_2O}^*$ is the partial pressure of the water vapor in the gas mix, if the gas is saturated at the given temp. of mixture.

$$\begin{array}{l} P_{H_2O} = 0.018 \text{ kPa} \\ P_{H_2O}^* = 0.035 \text{ kPa} \end{array} \rightarrow \% H_R = \frac{0.018}{0.035} \cdot 100\%$$

$$\therefore H_R = 51.43\%$$



(b) $H_{\text{final}} = 0.0115 \text{ kg H}_2\text{O} / \text{kg dry air}$, read after

locating the initial humidity point on the chart. ($H_i = 0.020$)

Since air is cooled down, we moved horizontally until we intersected the 100% saturation curve. Following the line to 15.6°C and then to 23.9°C we have found the final humidity value.

Finally, the percent humidity for the final reading is:

$$\% H_R = \frac{P_{\text{H}_2\text{O}}}{P_{\text{H}_2\text{O}}^*} ; P_{\text{H}_2\text{O}} = 0.018 \text{ kPa}$$

$$P_{\text{H}_2\text{O}}^* = 0.029 \text{ kPa}$$

$$\approx 0.030 @ T_f = 23.9$$

$$= \frac{0.018}{0.030} \cdot 100\% = (0.60) 100\%$$

$$\therefore H_R = 60\%$$

33.6-1 TIME FOR DRYING IN THE CONSTANT PERIOD

$$\text{From: } R = -\frac{L_S}{A} \frac{dX}{dt} \Rightarrow \Rightarrow t = \int_{x_2}^{x_1} dt \Rightarrow$$

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$$\Rightarrow t = \frac{L_S}{A} \int_{x_2}^{x_1} \frac{dX}{R_c}$$

$$\Rightarrow t = \frac{L_S}{AR_c} (X_1 - X_2)$$

$\left(\frac{L_S}{A}\right)_{\text{new}} = \left(\frac{L_S}{A}\right) \left(\frac{1}{4}\right)$, since we have increased the exposed area by 4 (both top & bottom), and we doubled the thickness of the wet solid.

$$\left(\frac{L_S}{A}\right)_{\text{new}} = \frac{24.4}{4} = 6.1 \frac{\text{kg dry solid}}{\text{m}^2 \text{ exposed surface}}$$

$$\Rightarrow t = \left(\frac{L_S}{A}\right)_{\text{new}} \frac{X_1 - X_2}{R_c} = (6.1) \left(\frac{0.45 - 0.20}{2.05}\right)$$

$$\Rightarrow t = \frac{6.1}{2.05} (0.25) = 0.746 \text{ h.}$$

$$\therefore t_{\text{new}} = 0.446 \text{ h.}$$

$$t_{\text{initial}} = \left(\frac{L_S}{A}\right)_{\text{initial}} \left(\frac{X_1 - X_2}{R_c}\right)_{\text{initial}} = \left(\frac{24.4}{2.05}\right) (0.55 - 0.22) = 3.93 \text{ h.}$$

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If we didn't change the dimension of the material, but we want to meet the

$$x_1 = 0.45$$

$$x_2 = 0.30$$

requirements, then the drying time in this case would have been:

$$\begin{aligned} t &= \frac{L_s}{A} \frac{1}{R_c} (x_1 - x_2) \\ &= \left(\frac{24.4}{2.05} \right) (0.45 - 0.30) = (11.9) (0.15) \\ &= 1.785 \text{ h.} \end{aligned}$$

33.6-3 Prediction in a Constant-Rate Drying Region

$$A = (0.61 \text{ m}) \times (0.61 \text{ m}) = 0.372 \text{ m}^2$$

$$b = 25.4 \text{ mm} = 0.0254 \text{ m}$$

$$v = 3.05 \text{ m/s}; L_s = 11.34 \text{ kg}$$

$$T_{ab} = 60^\circ \text{C}; T_{wb} = 29.4^\circ \text{C}$$

$$x_1 = 0.35; x_2 = 0.22$$

$$(a) R_c = \frac{h}{\lambda_w} (T - T_w) 3600 \quad (33.6-2)$$

From the steam table, we read:

$$H \approx 0.0141 \quad @ T = 60^\circ \text{C}$$

$\frac{\text{kg dry air}}{\text{kg } H_2O}$

Then the humid volume is:

$$V_H = \left[(2.83 \times 10^{-3}) + (4.56 \times 10^{-3}) H \right] (T + 273.15)$$

$$= \left[(2.83 \times 10^{-3}) + (4.56 \times 10^{-3}) 0.0141 \right] (60 + 273.15)$$

$$= 0.964 \frac{\text{m}^3}{\text{kg dry air}}$$

We want to calculate density, but we know that

$$m_H = (1.0 \text{ kg dry air}) + (0.0141 \text{ kg water})$$

$$\Rightarrow m_H = 1.0141 \text{ kg. Then:}$$

$$\rho = \frac{m_H}{V_H} = \frac{1.0141}{0.964} = 1.052 \frac{\text{kg}}{\text{m}^3}$$

Reading from the psychrometric table, we see that @ $T_{wb} = 29.4^\circ\text{C} \Rightarrow \lambda_w = 2432.14 \times 10^3$

Finally, we can calculate the drying rate:

$$R_c = \frac{h}{\lambda_w} (T - T_w)(3600); \quad \text{where } h = 0.0204 G^{0.8} \quad (336.9)$$

$$G = V \cdot \rho \cdot (3600) = (3.05)(1.052)(3600)$$

$$= 1155.96 \text{ kg/h-m}^2$$

$$\Rightarrow h = (0.0204)(1155.96)^{0.8} = 36.285 \text{ kg/h-m}^2$$

$$R_c = \left(\frac{36.29}{2432.14 \times 10^3} \right) (60 - 29.4)(3600)$$

$$\Rightarrow R_c = 1.6435 \frac{\text{kg}}{\text{h-m}^2}$$

Time needed to reach the desired drying rate is:

$$t = \frac{L_s}{A R_c} (X_1 - X_2)$$

$$\frac{L_s}{A} = \frac{11.34}{0.372} = 30.48 \Rightarrow$$

$$\Rightarrow t = \left(\frac{30.48}{1.6435} \right) (0.35 - 0.22) = (18.548) (0.13)$$

$$\therefore t = 2.41 \text{ h}$$

38.8-2 Drying when Radiation, Conduction, and Convection are Present

$$z_s = 25.4 \text{ mm} = 0.0254 \text{ m}$$

$$k_m = 43.3 \text{ W/m} \cdot \text{K}; k_s = 1.125 \text{ W/mK}$$

$$z_m = 1.59 \text{ mm} = 0.00159 \text{ m}$$

$$V = 3.05 \text{ m/s}; T = 60^\circ \text{C}; H = 0.010 \text{ kg H}_2\text{O/kg d.a.}$$

$$T_R = 104.4^\circ \text{C}; \epsilon = 0.94; R_c = ?; t = ?$$

Solution: $\rho = \frac{m}{V}; m = (1.0 \text{ kg}_{\text{dry air}}) + (0.010 \text{ kg H}_2\text{O}) = 1.010 \text{ kg}$

$$V_H = [(2.83 \times 10^{-3}) + (4.56 \times 10^{-3}) H] \cdot (T + 273.15)$$

$$= [(2.83 \times 10^{-3}) + (4.56 \times 10^{-3})(0.010)] (60 + 273.15)$$

$$= 0.958 \Rightarrow \rho = \frac{1.010}{0.958} = 1.054 \text{ kg/m}^3$$

$$\text{Then } q = v \cdot \rho (3600) = (3.05)(1.054)(3600) = 11575.99$$

$$\Rightarrow h_c = 0.0204 q^{0.8} = 0.0204 (11576)^{0.8} = 36.35 \frac{\text{kg/m}^2 \cdot \text{h}}{\text{kg/m}^2 \cdot \text{h}}$$

Reading the steam table, we see that
 at $T = 60^\circ\text{C}$ } $T_w = 28.89^\circ\text{C}$.
 $H = 0.010$

Since we must have that $T_s > T_w$, we take
 $T_s = 32.2^\circ\text{C}$.

$$\text{@ } T_s = 33^\circ\text{C} \Rightarrow \lambda_s = 2423.37 \text{ kJ/kg}$$

To predict h_R , we calculate:

$$h_R = 2(5.676) \frac{(T_R/100)^4 - (T_s/100)^4}{T_R - T_s} \quad (12.610)$$

$$= (0.94)(5.676) \frac{\left(\frac{104.4 + 273.15}{100}\right)^4 - \left(\frac{33 + 273.15}{100}\right)^4}{(104.4 + 273.15) - (33 + 273.15)}$$

$\Rightarrow h_R = 8.6188 \text{ W/m}^2\text{K}$. Then the overall heat transfer coefficient (conduction) is:

$$U_K = 1 / \left(1/h_c + z_M/k_M + z_s/k_s \right)$$

$$= 1 / \left(1/36.35 + \frac{0.00159}{43.3} + \frac{0.0254}{1.125} \right) = 19.95 \text{ W/m}^2\text{K}$$

Then the humid heat is:

$$C_s = 1.005 + 1.88 H = [1.005 + 1.88(0.010)] \times 10^3$$

$$= 1023.87 \text{ J/kg} \cdot \text{K}$$

This newly-obtained value can be substituted for $(h_c/k_r M_B)$ in the (33.8-9) equation. Thus:

$$\frac{(H_s - H) \lambda_s}{h_c/k_r M_B} = \left(1 + \frac{U_k}{h_c}\right)(T - T_s) + \frac{h_R}{h_c}(T_R - T_s) \quad (33.8-9)$$

$$\equiv \frac{(H_s - H) \lambda_s}{C_s} = \left(1 + \frac{U_k}{h_c}\right)(T - T_s) + \frac{h_R}{h_c}(T_R - T_s) \quad (*)$$

$$\Rightarrow \frac{(H_s - 0.010) \lambda_s}{1023.8} = \left(1 + \frac{19.95}{36.35}\right)(60 - T_s) + \left(\frac{8.6188}{36.35}\right)(104.4 - T_s)$$

For T_s assumed to be 32.2°C , $\lambda_s = 2424 \times 10^3 \text{ J/kg}$.

From the humidity chart for $T_s = 32.2^\circ\text{C}$, the saturation humidity $H_s = 0.031$. Substituting into the (33.8-9) eqn:

$$\frac{(0.031 - 0.010) (2424 \times 10^3)}{1023.8} = (1.5488)(60 - T_s) + (0.237)(104.4 - T_s)$$

$$\text{Solving for } T_s: T_s = \frac{67.9508}{1.7858} = 38.05^\circ\text{C}$$

For the second trial, assuming that $T_s = 32.5^\circ\text{C} \Rightarrow \left. \begin{array}{l} \lambda_s = 2423 \times 10^3 \\ H_s = 0.032 \end{array} \right\} \text{ from the Chart.}$

Substituting into (*) and solving for the unknown T_s :

$$\frac{(0.032 - 0.010) (2423 \times 10^3)}{1023.8} = \left(1 + \frac{19.95}{36.35}\right)(60 - T_s) + \left(\frac{8.6188}{36.35}\right)(104.4 - T_s)$$

$$\Rightarrow T_s = 36.74, \text{ assuming that } h_r \text{ does not change appreciably.}$$

Trial 3 : Assuming that
 $T_s = 32.7 \Rightarrow \lambda_s = 2424.08 \times 10^3 \text{ J/kg}$
 steam chart $H_s = 0.03$

Adjusting values for h_R :

$$h_R = E(5.676) \frac{\left(\frac{T_R}{100}\right)^4 - \left(\frac{T_s}{100}\right)^4}{T_R - T_s} = 0.94(5.676) \frac{\left(\frac{377.55}{100}\right)^4 - \left(\frac{305.25}{100}\right)^4}{377.55 - 305.25}$$

$$= 8.608 \text{ . Then using (33.8-9) eqn:}$$

$$\left(\frac{0.032 - 0.030}{1023.8}\right)(2424.084 \times 10^3) = \left(1 + \frac{19.95}{36.35}\right)(60 - T_s) + \left(\frac{8.608}{36.35}\right)(104.4 - T_s)$$

Given us: $T_s = 32.75$

Hence, the final value for T_s is taken to be 32.75.

Finally, calculating the drying rate:

$$R_c = \frac{Q}{A \lambda_s} = \frac{(h_c + U_k)(T - T_s) + h_R(T_R - T_s)}{\lambda_s} \quad (33.8-8)$$

$$= \frac{[(36.35 + 19.95)(60 - 32.75) + 8.608(104.4 - 32.75)]}{2424.08 \times 10^3}$$

$$\Rightarrow R_c = 3.1945 \text{ kg/h-m}^2$$

33.10-3. Through Circulation Drying in the Constant-Rate Period

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$$D_p = 12.7 \text{ mm} = 0.0127 \text{ m}$$

$$L = 63.5 \text{ mm} = 0.0635 \text{ m} \quad 355.35 \text{ K}$$

$$V = 0.914 \text{ m}^3/\text{s} @ T = 82.2^\circ \text{C} \quad \& \quad H = 0.01 \text{ kg H}_2\text{O}/\text{kg dry air}$$

$$\rho_{ds} = 1522 \text{ kg}/\text{m}^3; \quad \epsilon = 0.35$$

$$X_1 = 0.90$$

$$X_2 = 0.45 \quad \text{We want to find } t_{CRP}.$$

Time of drying at CRP:

$$t = \frac{\rho_s A_w X_c \ln(X_c/X)}{a \cdot w \cdot (T - T_w)} \quad (33.10-8)$$

$$= \frac{\rho_s X_c \ln(X_c/X)}{a \cdot K_y \cdot M_B (H_w - H)} = \frac{\rho_s A_w X_c (X_1 - X_c)}{Q_{cs} (T_1 - T_w) (1 - \exp(-\frac{h_{ax,1}}{Q_{cs}}))}$$

For the solid material, we set:

$$X = X_1 - X^* = 0.90 - 0.45 = 0.45$$

$$X_c = X_2 = 0.45$$

For gas mixture, we have:

$$T_1 = 82.2^\circ \text{C}$$

$$H_1 = 0.010 \frac{\text{kg H}_2\text{O}}{\text{kg dry air}}$$

From the steam tables:

$$T_w = 76^\circ \text{C}$$

$$H_w = 0.030$$

$$\lambda_w = 2637.45 \times 10^3$$

The density of the inlet air @ $T = 82.2^\circ \text{C}$ is:

$$\rho = \frac{m}{V}; \quad m = (1 \text{ kg dry air} + 0.010 \text{ kg water}) = 1.010$$

$$V_H = [(2.83 \times 10^{-3}) + (4.56 \times 10^{-3})(0.010)] (82.2 + 273.15)$$

$$= 1.021844 \text{ m}^3/\text{kg dry air}$$

$$\therefore C = \frac{1.010}{1.021844} = 0.9884087 \text{ kg dry air / h-m}$$

Then we can calculate the flux mass velocity of dry air

$$G = VE \left(\frac{1.0}{1.0 + 0.010} \right) = (0.914)(0.9884087)(3600) \left(\frac{1.0}{1.010} \right) = 3220.0595 \text{ kg dry air / h.m}^2$$

Since the inlet humidity $H_1 = 0.010$ and the outlet humidity H_2 will be less than $H_w = 0.030$, an approximate average H of 0.020 will be used in order to calculate the total average mass velocity. Thus, the approximate average G_t is:

$$G_t = G + G(H) = 3220.0595 + 3220.0595(0.020) = 3284.4607 \text{ kg air + H}_2\text{O / h.m}^2$$

To determine the surface area of bed:

$$a = G(1-E)/D_p \quad (33.10-15) \\ = 6(1-0.35)/0.0127 = 6(0.65)/0.0127 \\ = 3.9/0.0127 \Rightarrow a = 307.0866 \text{ m}^2 \text{ surface area / m}^3 \text{ bed}$$

Calculating the heat transfer coefficient: volume

$$h = 0.151 G_t^{0.59} / D_p^{0.41} \quad (33.10-13) \\ = (0.151) (3284.4607)^{0.59} / (0.0127)^{0.41} = 107.4 \\ \Rightarrow h = 107.4 \text{ W / m}^2 \cdot \text{K}$$

Then the average humid heat

$$c_s = 1.005 + 1.88 H \quad (33.3-6)$$

$$= 1.005 + 1.88(0.020) = 1.0426$$

$$\Rightarrow C_s = 1.0426 \times 10^3 \text{ J/kg} \cdot \text{K}$$

Calculating the bulk density of the bed ρ_s :

$$\epsilon = 1 - \rho_s / \rho \Rightarrow \rho_s = \rho(1 - \epsilon)$$

$$= (1522)(1 - 0.35) = 989.3$$

$$\Rightarrow \rho_{\text{bulk}} = 989.3 \text{ kg/m}^3$$

Finally, calculating the time required to dry the solid in the constant-rate period:

$$t = \frac{\rho_s \lambda_w x_1 (X_1 - X_c)}{G_{cs} (T_1 - T_w) (1 - e^{-\frac{h a x_1}{G_{cs}}})} =$$

$$= \frac{(989.3)(2637.45 \times 10^3)(0.0635)(0.90 - 0.45)}{(3220.0595)(1.0426 \times 10^3)(82.2 - 76.0) \left(1 - e^{-\frac{(107.7)(307)(0.0635)}{(3220)(1.0426 \times 10^3)}}\right)}$$

$$\Rightarrow t = 4806.948 \text{ sec} = 1.33526 \text{ hr.}$$